



smooth ride for trains on curves, as shown in Fig. 3.

After calculating the roll angle, the super elevation is obtained using the relation:

$$SE = \text{Track Gauge} \times \tan(\theta)$$

This calculation is implemented in the code around line 244 as:

```
// superelevation = TRACK_
GAUGE * tan(rollRad)
```

After completing the floating-point calculations, the final super elevation value is displayed in millimetres. Other sensor readings are displayed directly based on the selected mode.

Once calibrated, the system operates reliably and consistently. The system is fully self-contained, battery-powered, and compact, and can be housed in a small plastic or aluminium enclosure with appropriate openings for the sensors and OLED display. The track gauge input and temperature sensor are optional and may be connected using extension probes, if required.

In the Arduino IDE, select the Arduino Nano as the target board, choose the correct COM port, and upload the code. Fig. 4 shows the code snippet.

Optional improvements

The system may be enhanced with additional features. A Bluetooth-based wireless handheld display can be added for remote monitoring. An SD-card-based data logger can be incorporated to store and track parameters for later analysis. ESP-based controllers, such as ESP8266 or ESP32, can also be used to upload measurement data directly to Google Sheets or a cloud platform, enabling long-term monitoring and optimisation of track parameters.

Construction and testing

For construction, the source code is first uploaded to the Arduino Nano. After successful uploading, the circuit is assembled on a general-purpose PCB, as shown in the prototype in Fig. 1. Alternatively, the

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circuit may initially be assembled on a breadboard for testing and debugging. The I²C signal lines should be kept as short as possible to minimise noise and ensure reliable communication.

After assembly, the circuit is powered up, and the operation of individual modules is verified. The MPU6050 is calibrated to obtain a stable roll-angle reference, while the VL53L0X is tested for accurate distance measurement. The OLED display is checked for correct data visualisation before field use.

Following calibration, the system consistently displays super elevation, distance, and temperature readings. The compact, battery-powered design allows the unit to be housed in a small plastic or aluminium enclosure, making it suitable for on-site railway track measurements.

On power-up, the OLED display shows a start-up message, followed by the super elevation (SE) value. Pressing the push-button (SW1) cycles through the display modes. After two presses, the system displays the broad-gauge track width of 1676mm by adjusting the sensor position on the track.

Since rail temperature is a critical parameter for long welded rails, the system displays the track temperature using a DS18B20 sensor when the push-button is pressed for the third time. Pressing SW1 again repeats the display cycle. **EFY**



Kamal Das is an electronics tinkerer working as a Track Maintainer in Eastern Railway under SSE/PWay/TBAE, Howrah Division.



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